Number Theory: Subtraction

In this lab you've learned the basics of number theory as it relates to subtraction.

Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Names
Canon Miller
Zachary Munson

Summary

In this lab we practiced building a subtractor using both ones and twos compliment. Using both helped us to better understand the difference between ones and twos complement, and showed why twos complement is more common. We built the subtractors using a for loop to call our full adder from the previous lab.

Lab Questions

1 - Explain the differences between our Half Adder from last lab and the Half Subtractor from this lab. The half adder adds using XOR and has a carry out using A & B. The half subtractor only has an output of 1 when A = 1 and B = 0, and only has a borrow when the output is -1 / A = 0 and B = 1.

2 - What about the end around carry of One’s Complement makes it hard to use and implement? You have to output the last carry out to be the carry in for the first full adder which may cause an infinite loop. Our first solution was to create four adders and have the end around carry feed into the first adder, but it caused an error during synthesis due to the loop never closing.

3 - What is the edge case and problem with Two’s Complement number representation? The number range is not equal both ways like in ones complement. Instead of a range from -7 to +7 it is -8 to +7 which can cause negation issues.